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Genome-wide characterization and expression analysis of the *bHLH* gene family in response to abiotic stresses in *Zingiber officinale* Roscoe

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Abstract

Background The basic helix-loop-helix (bHLH) transcription factors play important physiological functions in the processes of plant growth, development, and response to abiotic stresses. However, a comprehensive genome-scale study of the ginger *bHLH* gene family has not been documented.

Results In this study, 142 *ZobHLH* genes were identified in the ginger genome. Using *Arabidopsis* bHLH proteins as a reference, *ZobHLH* genes were classified into 15 subfamilies and unevenly distributed on 11 chromosomes of ginger. Sequence characterization, multiple sequence alignment, phylogenetic analysis, conserved protein motifs and exon-intron distribution patterns were conducted to further analyze the evolutionary relationships among these *ZobHLH* proteins. The results of the duplicated event analysis demonstrated that the pivotal role of segment duplication in promoting the expansion of the *ZobHLH* gene family. Additionally, analysis of *cis*-regulatory elements as well as protein interaction networks indicated the potential involvement of ginger *ZobHLH* family proteins in plant growth and development, and response to adversity stress. RNA-seq and RT-qPCR results showed that *ZobHLH083* and *ZobHLH108* play key roles in response to salt stress and waterlogging stress, respectively.

Conclusion In this study, we systematically analyzed the characteristics of *ZobHLH* proteins in ginger, discovering that these genes play critical roles in ginger rhizome expansion and response to salt and waterlogging stresses. The present study provides a theoretical foundation for the further research on *ZobHLHs* and will help to explore the functional properties of *ZobHLH* genes.

Keywords *bHLH*, Ginger, Transcription factors, Rhizome development, Abiotic stress

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Background

Adverse factors such as heavy metals, drought, high salt, or extreme temperatures, which significantly disrupt plant normal physiological activities, reducing production and quality [1]. However, plants have evolved complex signal sensing and defense systems when exposed to adversity. During this process, transcription factors (TFs) play critical roles and mainly function to regulate relative genes expression and then modify plant growth, physiology and metabolic pathways, and then enhance the resistance to abiotic stresses [2]. For example, MYB TFs can bind to MYB-binding elements in the promoter regions of various functional genes to activate the expression level of stress-responsive genes and then initiate plant response to adversity [3]. The bHLH family is one of the largest TF families in plants and plays an important role in plant response to abiotic stresses including waterlogging and high salt stress [4].

The bHLH TFs were first identified in animals and later in the majority of eukaryotes, deriving its name from the presence of the bHLH (basic Helix-Loop-Helix) domain, which spans approximately 60 amino acids [5, 6]. The N-terminus features an α -helix within the basic structural domain, comprising approximately 15–20 amino acids and enriched with basic residues binding to the *cis*-elements. The HLH region is located to the C-terminal, containing two α -helices connected by a relatively poorly conserved loop, and relies on the interaction of hydrophobic amino acids to form homo- or heterodimers [7, 8]. There are 15 to 32 subfamilies according to the taxonomy of bHLH proteins [9, 10]. Most bHLH TFs are crucial for plant growth and metabolism, influencing secondary metabolism, light morphogenesis and signal transduction [11]. For example, maize anthocyanins are regulated by the highly conserved MYB-bHLH-WD40 transcriptional regulatory complex [12]. In *Fragaria ananassa*, the genes *FabHLH25*, *FabHLH29*, *FabHLH80*, and *FabHLH98* are involved in fruit anthocyanins production and hormone signal transduction [13]. The TF *AabHLH112* enhances the production of sesquiterpenes in *Artemisia annua* [14]. Conversely, silencing of the genes *SmMYC2a* and *SmMYC2b* results in reduced accumulation of tanshinone and phenolic acids [15]. In addition, bHLH TFs play important roles in plant stress response. The pepper gene *CabHLH035* promotes salt tolerance by regulating ion homeostasis and proline biosynthesis [16]. Overexpressing *SlbHLH22* and *SlbHLH96* in tomato enhances drought and salt tolerance [17, 18]. *OsbHLH057* activates the expression of the *Os2H16* gene, thereby promoting disease resistance and drought tolerance in rice [19]. Moreover, bHLH proteins regulate tryptophan biosynthesis and various hormones, significantly contributing to hormone-mediated regulatory networks

[10, 20]. In *Arabidopsis*, *bHLH3*, *bHLH13*, *bHLH14* and *bHLH17* function as transcriptional repressors affecting responses to jasmonic acid (JA) [21]. In sweet sorghum, *SbbHLH85* plays a crucial role in root development and mediates the response to salt stress by regulating abscisic acid (ABA) and auxin signaling pathways [22]. Therefore, bHLH TFs are integral to plant development and stress response. However, studies on the function and physicochemical properties of the *ZobHLH* genes in *Zingiber officinale* Roscoe are lacking.

Ginger, widely utilized for its medicinal and culinary properties, is globally cultivated primarily for its underground rhizome [23]. Ginger has great economic value, but it is susceptible to various detrimental factors, such as salt [24], drought [25], waterlogging [26], and high temperature [27]. Recent studies have shown that salinity causes a decrease in photosynthetic capacity of ginger, and plant dwarfing [28]. Waterlogging inhibits the root vitality and hampers its growth and development [29]. Both of them lead to a decrease in the accumulation of minerals in ginger, which in turn cause a significant reduction in its yield. Uncovering the mechanisms underlying growth, development and tolerance to abiotic stresses can facilitate the selection and breeding of resilient and high-yielding ginger cultivars. Recent advancements in sequencing technology and the completion of the ginger genome sequence have enabled the identification of gene families involved in stress responses on a genome-wide scale [30]. This study employs bioinformatics to identify *ZobHLH* family genes in ginger, analyzing their physicochemical properties, gene structure, conserved motifs, phylogenetic relationships and gene expression patterns, which will elucidate the roles of *ZobHLHs* in abiotic stress responses. These results will provide a global perspective for the study of evolution and functional characterization of the *ZobHLHs* in ginger, as well as further research on the function of *ZobHLHs* in growth and development and the mechanism of stress resistance, and also provide a theoretical reference for the breeding of stress-tolerant ginger varieties.

Materials and methods

Identification of the *ZobHLH* gene family

The ginger bHLH family members were identified by Hidden Markov Modelling (HMMER) and the ginger genome has been published online [30]. The HMMER profile for HLH (PF00010) was downloaded from the Pfam database (<https://pfam.xfam.org/>), and the candidate genes were obtained from the ginger genome using HMMER software with an e-value cutoff set to 10^{-5} . The protein sequences of the identified genes were subsequently analyzed for redundant and structurally incomplete domain sequences using NCBI_CDD (<https://www.ncbi.nlm.nih.gov/Structure/cdd/>).

ncbi.nlm.nih.gov/Structure) and SMART (<http://smart.embl-heidelberg.de/>). The genes were then renamed sequentially based on their chromosomal positions, and the physicochemical properties of ZobHLH proteins predicted by ExPASy (<https://web.expasy.org/protparam/>).

Sequence alignment and phylogenetic relationship, gene structure, and conserved motif analyses of the *ZobHLHs*

Multiple sequence alignment of *ZobHLH* protein sequences was performed with ClustalW2, and phylogenetic trees were constructed using MEGA 11.0 software by Neighbor-joining (NJ) with 1000 bootstrap replicates [31]. The bHLH protein sequences of *Arabidopsis* were downloaded from the PlantTFDB website (<https://planttfdb.gao-lab.org/>). Conserved structural domains were identified using the WEB LOGO tool (<http://weblogo.berkeley.edu/logo.cgi>) to generate sequence logos. The exon-intron structure of *ZobHLHs* was obtained from the gene annotation files of the ginger genome and visualized by the online software gene structure display server (<http://gsds.cbi.pku.edu.cn/>). The conserved motifs of *ZobHLH* proteins were identified by MEME v5.1.1 (<http://meme-suite.org/tools/meme>), allowing for a maximum of 20 and optimal motif widths ranging from 6 to 50.

Chromosomal distribution, gene duplication and synteny of the *ZobHLHs*

The genomic location of *ZobHLHs* was mapped to their respective chromosomes using an annotation file from the ginger genome. Gene duplication events within *ZobHLHs* were analyzed using the Multiple Co-linear Scanning Toolkit (MCScanX) with default parameters [32]. Comparative collinearity analyses between ginger and other species (including *Arabidopsis*, rice, banana, maize, tomato) were detected and visualized using TBtools v1.089 [33]. Genome sequences and annotation files of *Arabidopsis*, rice, banana, maize, and tomato were downloaded from EnsemblPlants (<https://plants.ensembl.org/index.html>). Ka/Ks analyses were performed to evaluate selection pressure by the online software PAL2NAL (<https://www.bork.embl.de/pal2nal/>).

Analyses of *cis*-regulatory elements and prediction of the protein-protein interaction network of the *ZobHLHs*

Sequence located 2000 bp (base pairs) upstream of the *ZobHLHs* genes were extracted and analyzed for *cis*-acting elements using PlantCARE (<https://bioinformatics.psb.ugent.be/webtools/plantcare/html/>). Furthermore, the STRING database (<https://string-db.org/>) was employed for the interaction networks and a confidence

threshold was set to 0.4. The network was visualized by Cytoscape software (v3.10.1).

GO (gene ontology) functional annotation and enrichment analysis of *ZobHLHs*

The *ZobHLH* protein sequences were annotated using the Blast2GO program to assign the GO terms (<https://amigo.geneontology.org/amigo/term/>), and the e-value for GO analysis was 1.0e-6.

Plant materials and expression analysis of *ZobHLHs* by RNA-seq and RT-qPCR

The ginger species utilized in this study was 'Zhugen' ginger. Initially, the gingers were germinated under controlled conditions with a 24 h dark photoperiod at 26 °C and 70% relative humidity. Upon budding, healthy ginger buds were selected and cultivated in the field, and samples were collected at five stages of rhizome expansion (R1 ~ Rh5), respectively [34]. Simultaneously, another ginger buds were transplanted into pots with a diameter of 20 cm and grown in mixed substrate including garden soil, cocopeat, and perlite in a ratio of 6:3:1 under conditions of 26 °C, approximately 10,000 lx light intensity, 70–80% humidity, and 12-hour light-dark photoperiod.

After two months of routine cultivation and management, ginger seedlings with uniform and robust growth were selected for stress treatments. For salt stress, the plants were watered with 20 g L⁻¹ of NaCl solution for three consecutive days. For waterlogging stress, the double-jacketed pot method [29] was employed to maintain soil at a supersaturated level, with periodic rehydration until the water surface was 3 cm above the soil surface. Samples including leaves and roots under salt and waterlogging stress treatments for 3 days were collected, respectively, and then rapidly snap-freezing in liquid nitrogen for RNA-seq analysis and RT-qPCR analyses.

Total RNA was extracted according to the instruction of FastPure Universal Plant Total RNA Isolation Kit (Vazyme), and cDNA synthesis was performed based on HiScript II Q RT SuperMix for qPCR (+gDNA wiper) (Vazyme). RT-qPCR analysis was conducted using Taq Pro Universal SYBR qPCR Master Mix (Vazyme) on a CFX Connect Fluorescent Quantitative PCR Detection System (Bio-Rad Laboratories). Each 20 µL reaction mixture contained 10 µL Taq Pro Universal SYBR qPCR Master Mix, 0.8 µL primer (10 µM Forward and Reverse), 6.7 µL ddH₂O, and 2.5 µL cDNA template. The amplification program for RT-qPCR run as follows: initial denaturation at 95 °C for 3 min, followed by 44 cycles of 95 °C for 30 s and 60 °C for 30 s, with a melt curve analysis from 60 to 95 °C. DEGs (Differently expressed genes) with FPKM (fragments per kilobase of transcript

per million fragments mapped) values greater than 15 and $\log_2(\text{FC})$ greater than 1.5 were selected for quantification of leaves and roots under salt and waterlogging stress, and a total of 17 genes were screened, namely, *ZobHLH003/018/020/032/043/065/083/086/088/095/108/111/122/126/128/131/142*. The reference gene *RBP* [35] was used for normalization, and each experiment included three biological replicates. Relative expression levels of *ZobHLHs* were calculated by $2^{-\Delta\Delta\text{CT}}$ method [36]. Primer information for RT-qPCR were listed in Supplementary Table S1.

Subcellular location analyses

Predictions for subcellular localization of *ZobHLH* proteins were conducted using WOLF PSORT69 (<https://www.genscript.com/psort.html>). To further determine the location of the ginger *ZobHLH* genes, *ZobHLH083* and *ZobHLH108* which did not contain a stop codon, were amplified by PCR with a pair of specific primers (Supplementary Table S2), and the entire coding regions of *ZobHLH083* and *ZobHLH108* were constructed into the pCambia1302 vector which contained a GFP protein under the control of CaMV 35 S, respectively. Subsequently, the recombinant plasmids were transferred into *Agrobacterium tumefaciens* GV3101, then injected into the epidermal leaves of tobacco for 48 h. The GFP signal can be observed under a laser confocal microscope (Olympus FV 1200).

Results

Identification and physicochemical properties of the *ZobHLHs*

After removing redundancy, a total of 142 *ZobHLH* protein sequences containing bHLH conserved structure domains were identified. All the *ZobHLH* protein sequences were listed in Supplementary Table S3. *ZobHLH* genes were mapped to 11 chromosomes and sequentially renamed as *ZobHLH001~ZobHLH142* based on their chromosomes position (Supplementary Table S4).

The physicochemical properties of *ZobHLH* proteins varied widely (Supplementary Table S5). According to the prediction of ExPASy, these proteins ranged in length from 95 to 1134 amino acids, with molecular masses between 10615.09 and 126549.30 Da, and theoretical isoelectric points (pI) from 4.80 to 9.87. Among them, 96 *ZobHLH* proteins were acidic ($\text{pI} < 7$), and 46 were classified as basic ($\text{pI} > 7$). The protein with the lowest instability index was *ZobHLH020* (42.44), while the highest was *ZobHLH079* (78.86). All the proteins of instability indices were greater than 40, indicating *ZobHLH* belong to unstable proteins. The aliphatic index ranged from 53.75 (*ZobHLH123*) to 96.28 (*ZobHLH034*). The grand

average of hydropathicity (GRAVY) was negative, showing that *ZobHLHs* were hydrophilicity (Supplementary Table S4). Subcellular localization predictions revealed that 123 *ZobHLH* proteins were localized in the nucleus, six in the chloroplast, five in the cytoplasm, three in the mitochondrion, two in both the peroxisome and plasma membrane, and one in the endoplasmic reticulum (Supplementary Table S6).

Multiple sequence alignment, phylogenetic analysis, and classification of *ZobHLHs*

According to multiple sequence comparison, all the *ZobHLHs* contain a conserved basic region and two helical regions (Fig. 1). Notably, residues Glu-4, Arg-5, Arg-9, and Arg-10 in the basic region, Leu-20 and Pro-24 in helix 1, and Asp-58, Ala-60, Leu-66, Leu-76 in helix 2 show high conservation ($> 75\%$). Remarkably, residues Arg-9, Arg-10, Leu-20, Pro-26, and Leu-76 demonstrate a consistency level exceeding 95%, highlighting their critical roles in *ZobHLHs* function.

To analyze the evolutionary relationships of ginger *ZobHLH* gene family, 142 *ZobHLH* proteins were aligned with 153 *A. thaliana* bHLH proteins using ClustalW2. As shown in Fig. 2 and 295 bHLH proteins were categorized into 15 distinct subfamilies (A-O), which consistent with the classification results proposed by Wang et al. [37]. *ZobHLHs* were distributed in all subfamilies, subfamily A contains 23 proteins, while subfamily G includes only one *ZobHLH* protein (Fig. 2).

Gene structure, and conserved motifs of the *ZobHLHs*

Analysis of conserved motifs among *ZobHLH* proteins revealed variations both in the type and number. As shown in Fig. 3B, all the *ZobHLH* proteins contained highly conserved Motif-1 and Motif-2. While Motif-15 existed exclusively in 11 members of subfamily N. *ZobHLHs* in same subfamily usually have similar motif compositions. For example, genes in subfamily M typically contained Motif-1, Motif-2, and Motif-3, while members of the subfamily O contained Motif-14 in addition to the motifs found in subfamily M. Interestingly, some conserved motifs were found to occupy specific positions in the protein sequence. Motif-9 and Motif-10 were invariably located at the N-terminus, while Motif-4, Motif-8, Motif-5, Motif-13 and Motif-15 were consistently located at the C-terminus. Further analysis indicates that all the proteins contain the bHLH characteristic domain (Fig. 3C), some contain additional domains, such as a reticulon domain in *ZobHLH039*, and a ZnF_C2HC structure domain in *ZobHLH062*. For the gene structure, we found that 142 *ZobHLH* genes exhibit exon counts ranging from one to 11 (Fig. 3D). Except from 13

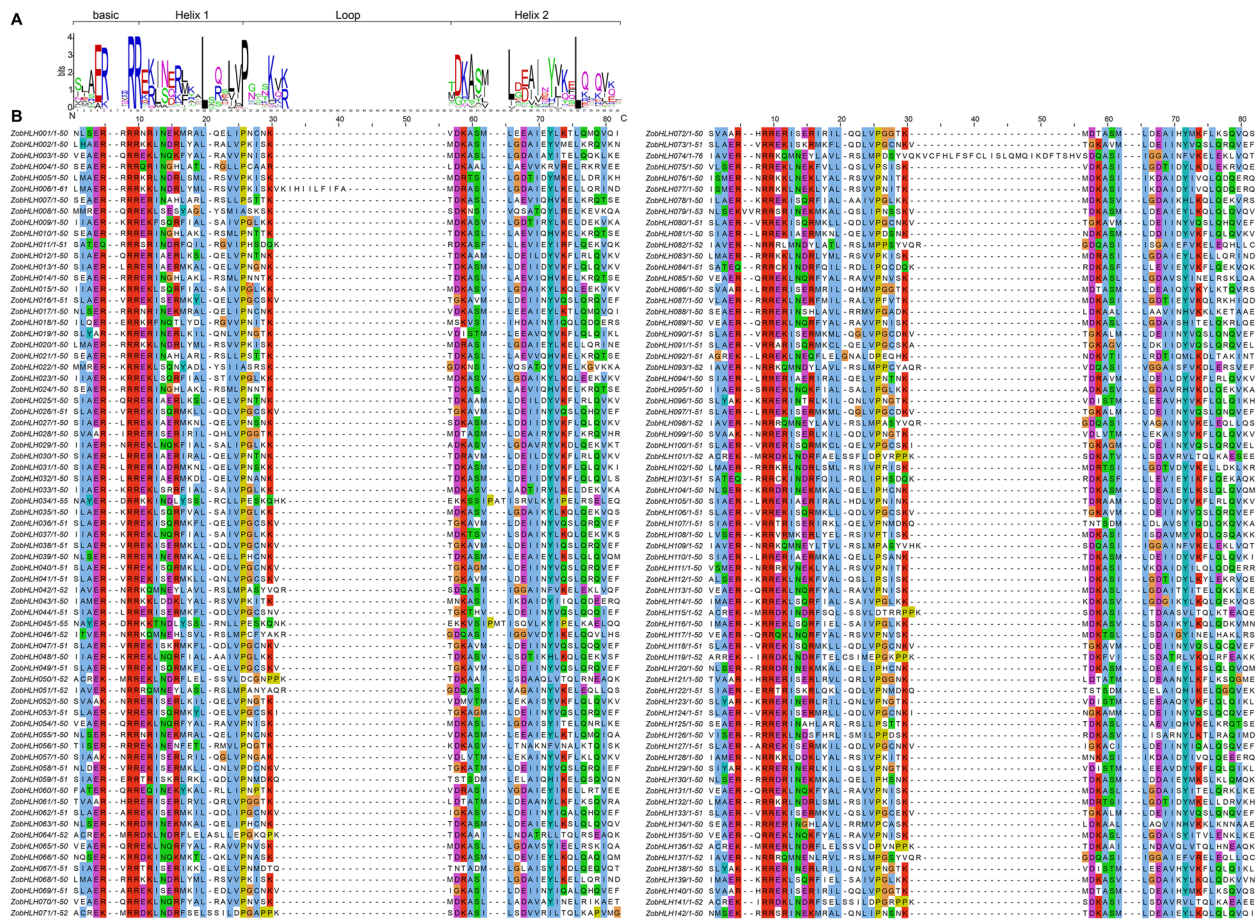


Fig. 1 Multiple sequence comparison of structural domains of ZohHLH proteins. **A** Sequence logo for the ZohHLH domains. The overall height of each stack represents the conservation of the sequence at that position. **B** Different color indicates the degree of amino acid identity at each position

ZohHLHs that lack introns, the CDSs of the remaining genes are interrupted by introns, and the intron numbers ranging from one to 10. Notably, eight of the intron-lacking genes belong to subfamily C, while five belong to subfamily I. Despite differences in exon locations, genes in the same subfamily generally exhibit similar exon numbers and distribution patterns.

Chromosomal distribution, evolutionary analyses of the *ZohHLHs* within and between species

As was shown in Fig. 4A, the distribution of *ZohHLH* genes across the chromosomes was uneven. For example, Chr04 contained eight genes, Chr06 contained 19 genes, and the remaining chromosomes contained between 10 and 17 *ZohHLHs*. In this study, A tandem duplication of genes (*ZohHLH076/077*) on Chr06 was detected (Fig. 4A). Additionally, 42 pairs of segment duplication genes were detected, involving 74 *ZohHLH* genes, in their respective subfamilies (Fig. 4B). To elucidate the evolutionary pressures influencing the *bHLH* gene family, the

ratios of KA/KS for *ZohHLH* gene pairs were calculated (Supplementary Table S7). The results showed that all the homologous gene pairs have KA/KS values less than 1, suggesting that purifying selection has strongly shaped the evolution of *ZohHLH* genes.

To further elucidate the evolutionary relationship of the *bHLH* genes across different species, a collinearity map comparing ginger with six other species was constructed using TBtools (Fig. 5, Supplementary Table S8). The results showed that no collinear genes existed between ginger and *Arabidopsis*, while seven pairs of collinear genes were identified in tomato. In contrast, 16 pairs of collinear genes were found in rice, 19 pairs in maize, 43 pairs in wheat, and 184 pairs in banana. These results demonstrate that ginger shares a closer evolutionary relationship with monocotyledons compared to dicotyledons.

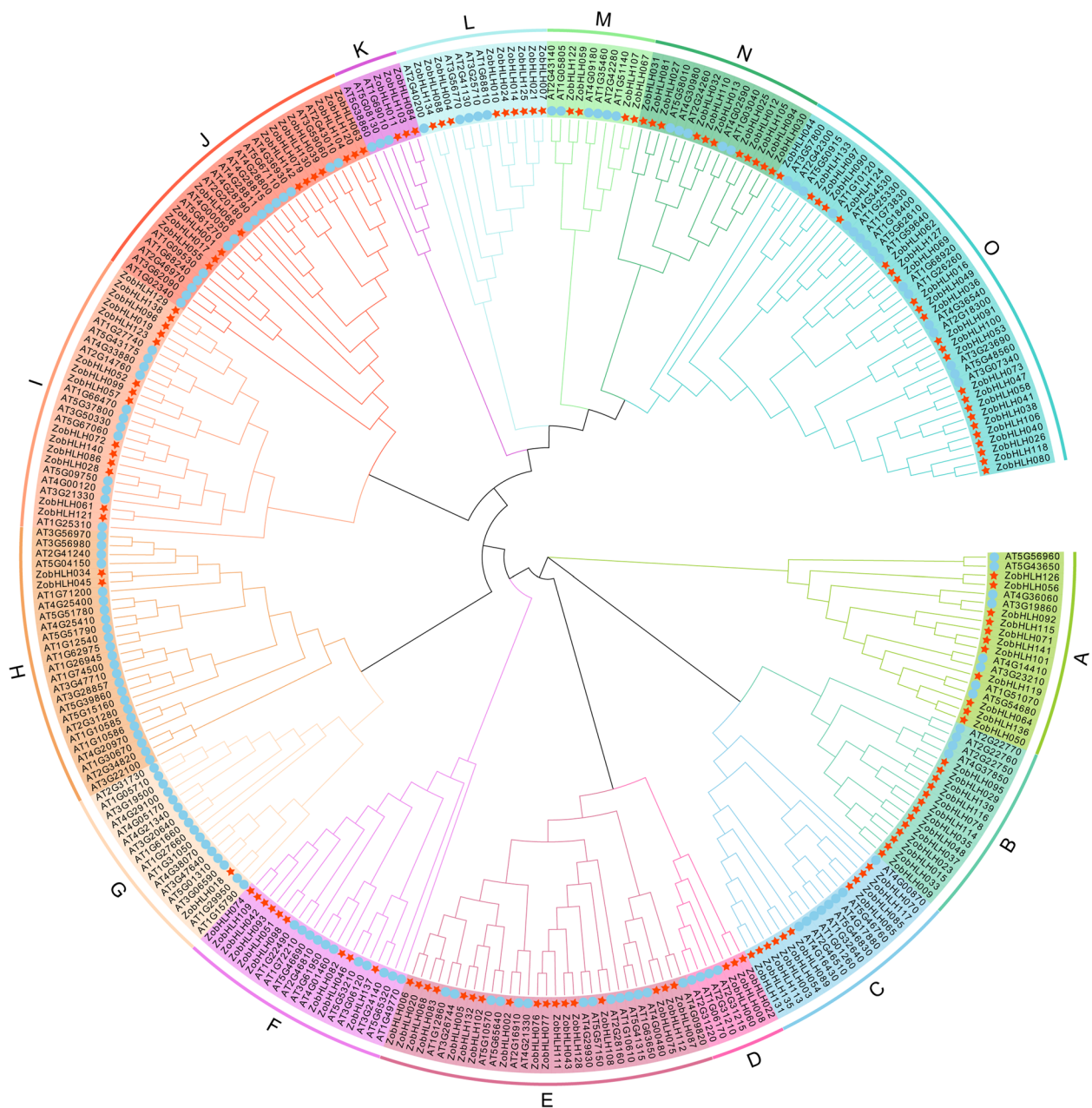


Fig. 2 Phylogenetic tree of *bHLH* TFs in ginger and *Arabidopsis*. Different colors indicate different subfamilies, red stars indicate ginger, skyblue circles represent *Arabidopsis*

Cis-element analyses of the *ZobHLHs*

To determine the type and distribution of *cis*-acting elements in the *ZobHLH* promoter, we extracted 2000 bp upstream of the *ZobHLHs* using PlantCARE. As showed in Fig. 6 and Supplementary Table S9, the predominant *cis*-elements are associated with hormone response, including auxin, abscisic acid, gibberellins, methyl

jasmonate, salicylic acid, et al. Additionally, several *cis*-elements related to plant growth and development were identified, such as *cis*-elements involved in endosperm expression, meristem expression, and flavonoid biosynthetic genes regulation. Notably, some *cis*-elements related to stress responses, including defense and stress responsiveness, like MYB binding site involved in drought-inducibility, wound responses, anoxic specific inducibility, low-temperature responsiveness and other



Fig. 3 Conserved motifs, gene structures and phylogenetic relationships of ZobHLHs. **A** Phylogenetic tree of 142 ZobHLH proteins. **B** Conserved motifs in ZobHLHs, the conserved motifs are represented in different colors, black lines indicate introns. **C** Structural domains contained in ZobHLHs, green indicates the bHLH structure domain, yellow indicates the bHLH-MYC_N structure domain, red indicates the reticulon structure domain and dark green indicates the ZnF_C2HC structure domain. **D** Exon-intron structural domains of the ZobHLHs, yellow boxes indicate exons, green boxes indicate 5'- and 3'-untranslated regions

abiotic stress-related components. In Fig. 6B, among these three types of components (hormone, growth and stress responsiveness), the number of components related to abiotic stress was the highest, indicating that *ZobHLH* genes played important roles in responding to abiotic stress.

Prediction of the *ZobHLHs* protein-protein interaction network

The prediction of protein-protein interaction revealed significant interactions between the *ZobHLH* proteins

(Fig. 7), which could be categorized into three layers based on their connectivity [23]. The innermost circle, characterized by the highest connectivity (> 15), comprises four members, namely *ZobHLH103*, *ZobHLH126*, *ZobHLH132* and *ZobHLH141*. The second circle contains 12 *ZobHLHs*, and the outermost circle contains 30 *ZobHLH* proteins with connectivity less than nine. Among them, *ZobHLH103* is homologous to *Arabidopsis* BIM2, *ZobHLH126* is homologous to *AtbHLH041*, *ZobHLH132* is homologous to *AtbHLH93*, and *ZobHLH141* is a homologous protein of *AtbHLH115*.

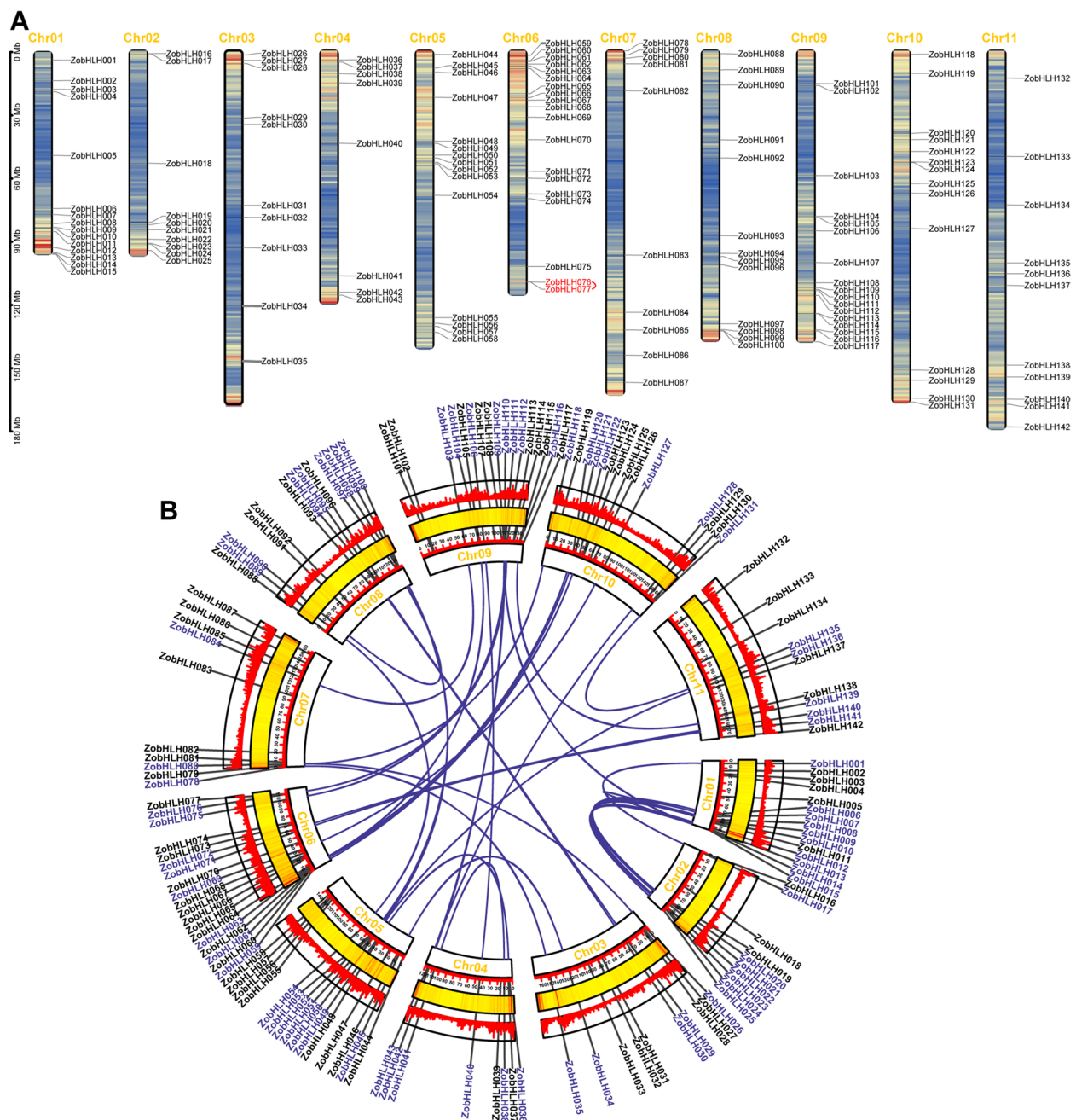


Fig. 4 Chromosomal distribution and segmental duplication collinearity analysis. **A** Chromosome distributions of *ZoblHLH*s. All chromosomes have the *ZoblHLH* genes and the red arcs connect the tandem duplication genes. **B** Chromosomal distribution and segmental duplication collinearity map of the *ZoblHLH*s in ginger. Blue font and blue lines indicate segmental duplication gene pairs in the ginger genome

BIM2 is important in brassinosteroid signaling [38], *AtbHLH93* is involved in stomatal defense and cell differentiation [39], *AtbHLH041* is associated with the signaling of auxin and cytokinin [40], *AtbHLH115* plays an important role in iron homeostasis and adversity regulation [41]. Hence, we can speculate that *ZoblHLH* proteins may function in several aspects of plant growth and

development, including hormone signaling, cell differentiation, and adversity response.

GO analysis of *ZoblHLH*s proteins

To further elucidate the functions of *ZoblHLH*s genes, we conducted gene ontology annotation analysis. The GO annotation of *ZoblHLH* includes biological processes,



Fig. 5 Synteny analysis of *bHLH* genes between ginger and five plant species (*A. thaliana*, *Solanum lycopersicum*, *Oryza sativa*, *Zea mays*, *Triticum aestivum*, and *Musa acuminata*). The grey line shows the collinearity of ginger with the other species, while the red line highlights the collinearity *bHLH* gene pairs

molecular function and cellular component (Supplementary Table S10). In particular, ZobHLH proteins are

involved in a variety of biological processes, including the regulation of biological processes, intracellular metabolic

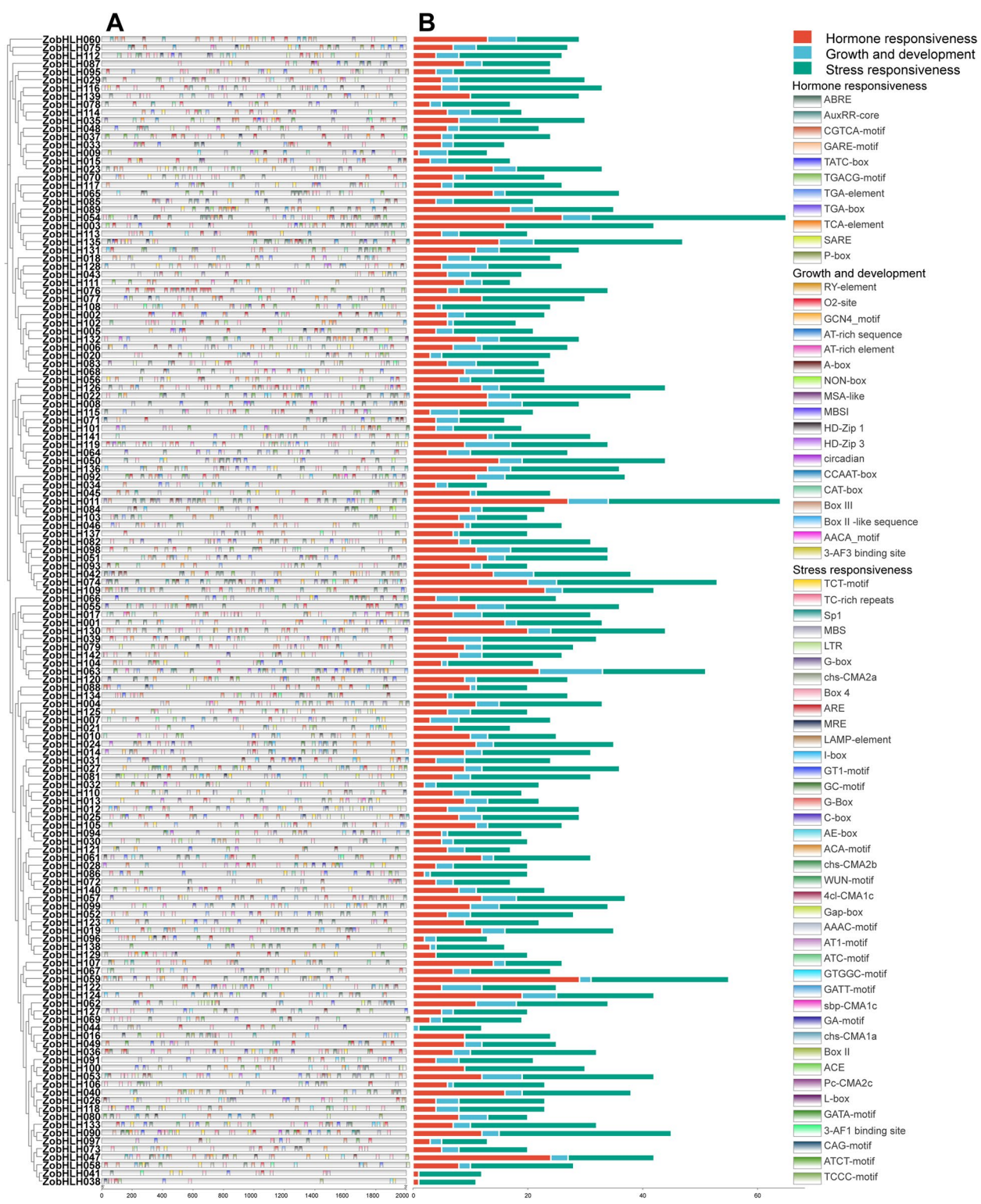


Fig. 6 *Cis*-regulatory elements of the *ZobHLH* promoters. **A** Distribution of *cis*-acting elements in the 2000 bp sequence upstream of the 142 *ZobHLH* genes, with different color blocks indicating different acting elements, and some elements overlapping. **B** Histogram showing the proportion of various *cis*-elements, different colors indicate different *cis*-element types

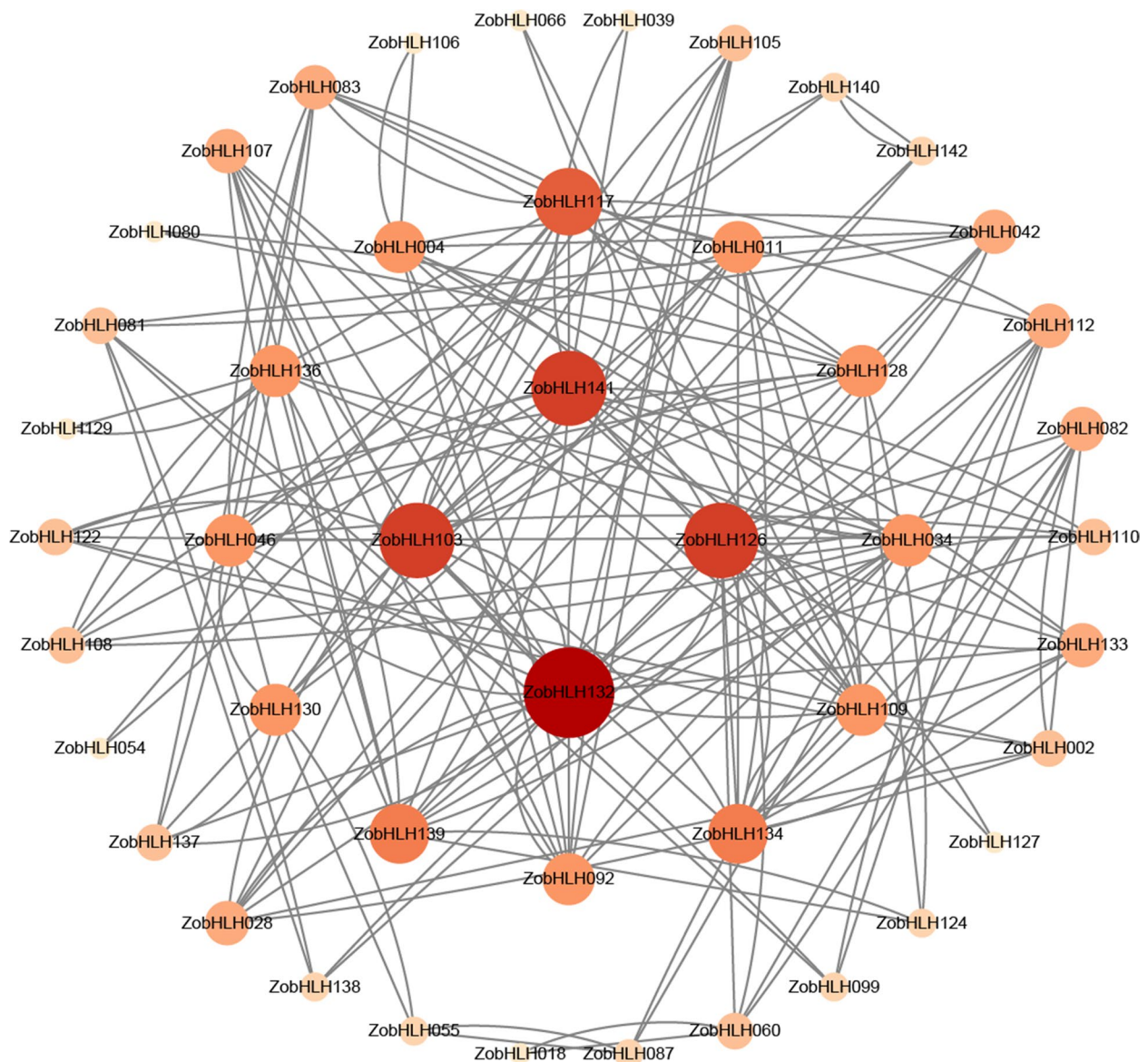


Fig. 7 Interaction network of ZobHLH proteins. Grey line connections indicate predicted interactions between the proteins, and the nodes (red circles in the figure) range from large to small and from dark to light red, indicating a gradual decrease in the number of interacting proteins

processes, regulation of DNA-templated transcription, anatomical structure development and response to abiotic stimulus. Among the molecular functions, ZobHLH proteins are involved in DNA-binding transcription activity, and DNA binding. In the cellular component, ZobHLH proteins are mainly involved in intracellular membrane-bounded organelle, membrane-bounded organelle.

Expression profiles of the *ZobHLHs* in ginger at different developmental stages

To analyze the expression profiles of *ZobHLHs* during 'Zhugen' ginger rhizome expansion, we determined transcriptome data from five developmental stages (Supplementary Table S11, Fig. 8A). At the Rh1 stage, 42 *ZobHLH* genes were significantly increased ($\text{FPKM} \geq 5$). There are 37, 38, 41, and 57 *ZobHLHs* upregulated in Rh2, Rh3, Rh4, and Rh5 stages, respectively. Interestingly, 34 genes were highly expressed at all five expansion periods simultaneously, such as *ZobHLH065*, *ZobHLH064*, et al.

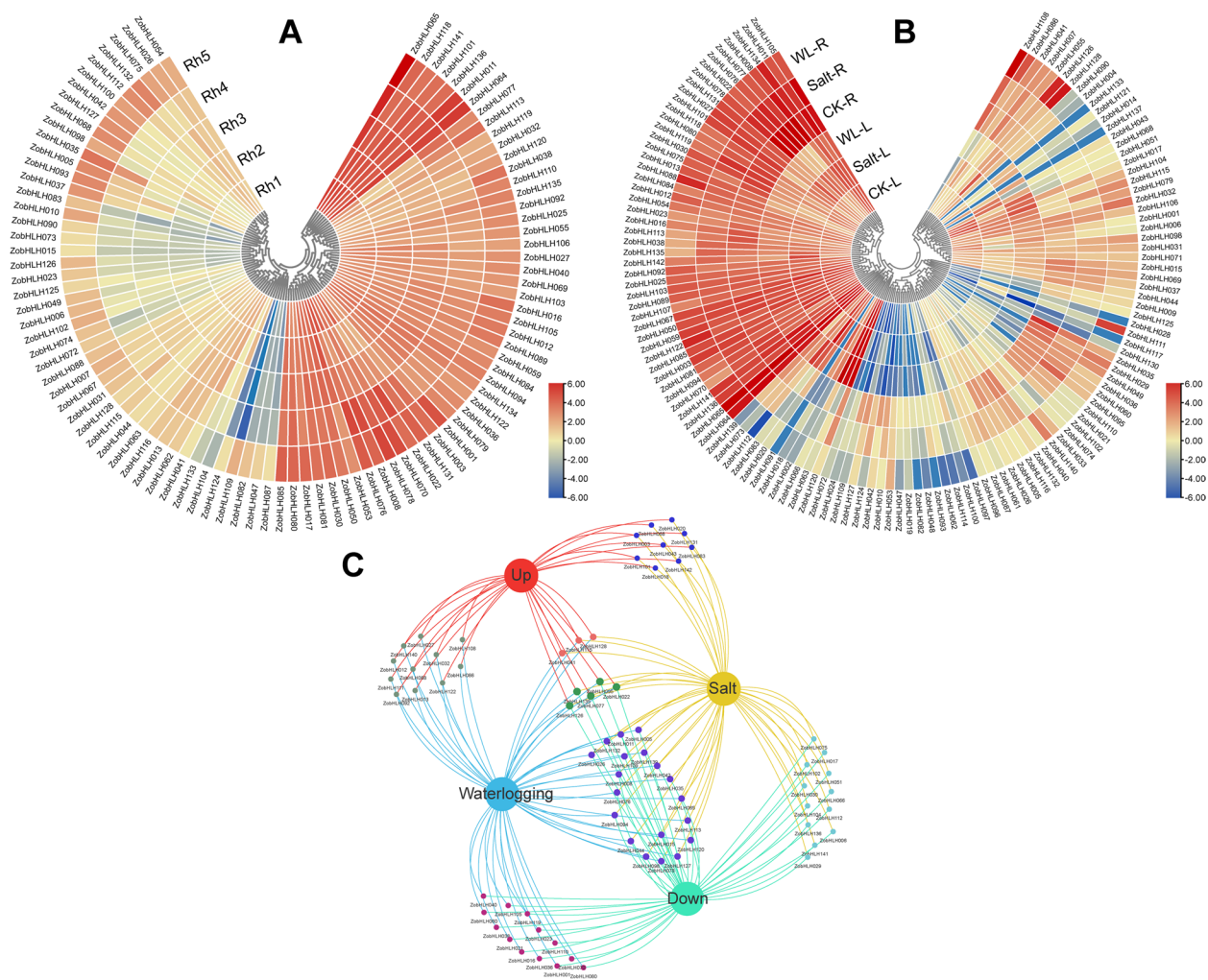


Fig. 8 Expression profiles of the *ZoblHLHs* in ginger at different developmental stages and abiotic stresses. **A** Heat map of *ZoblHLH* genes expression during five developmental periods of ginger rhizome expansion. **B** Expression profiles of *ZoblHLH* genes in leaves and roots of ginger under salt stress and waterlogging stress. Log₂FPKM normalization was used, and color blocks from blue to red indicate low to high gene expression. **C** Venn diagram of DEGs up-regulated and down-regulated in leaves and roots under salt and waterlogging stresses

The expression level of *ZoblHLHs* in ginger under salt and waterlogging treatments

In this study, we performed RNA-seq analysis on different parts of ‘Zhugen’ ginger subjected to salt and waterlogging stresses, respectively (Supplementary Table S12, Fig. 8B). Under salt stress, 48 genes showed significant differential expression, with 28 in leaves (15 up-regulated) and 25 in roots (three up-regulated). Four genes (*ZoblHLH142*, *ZoblHLH022*, *ZoblHLH139*, and *ZoblHLH077*) were differentially expressed in both leaves and roots, among which *ZoblHLH142* up-regulated in all tissues. Also in leaves, *ZoblHLH083* was significantly up-regulated and had the highest expression level. After waterlogging treatment, 53 genes were differentially expressed in roots, of which 15 genes were up-regulated and 35 genes were down-regulated.

Only two genes were down-regulated in leaves. Among the up-regulated genes, *ZoblHLH108* had the highest expression level, which was more than 10 folds of the control. In addition, a total of 27 genes were differentially expressed in both salt and waterlogging stresses, such as *ZoblHLH128*, *ZoblHLH041*, and *ZoblHLH115* were significantly up-regulated (Fig. 8C).

RT-qPCR

Additionally, 17 DEGs was selected for validation by RT-qPCR. Under salt stress, *ZoblHLH083*, *ZoblHLH131*, *ZoblHLH043*, *ZoblHLH018*, *ZoblHLH020*, and *ZoblHLH065* were significantly up-regulated in leaves (Fig. 9). *ZoblHLH142* and *ZoblHLH003* were up-regulated in both leaves and roots. Under

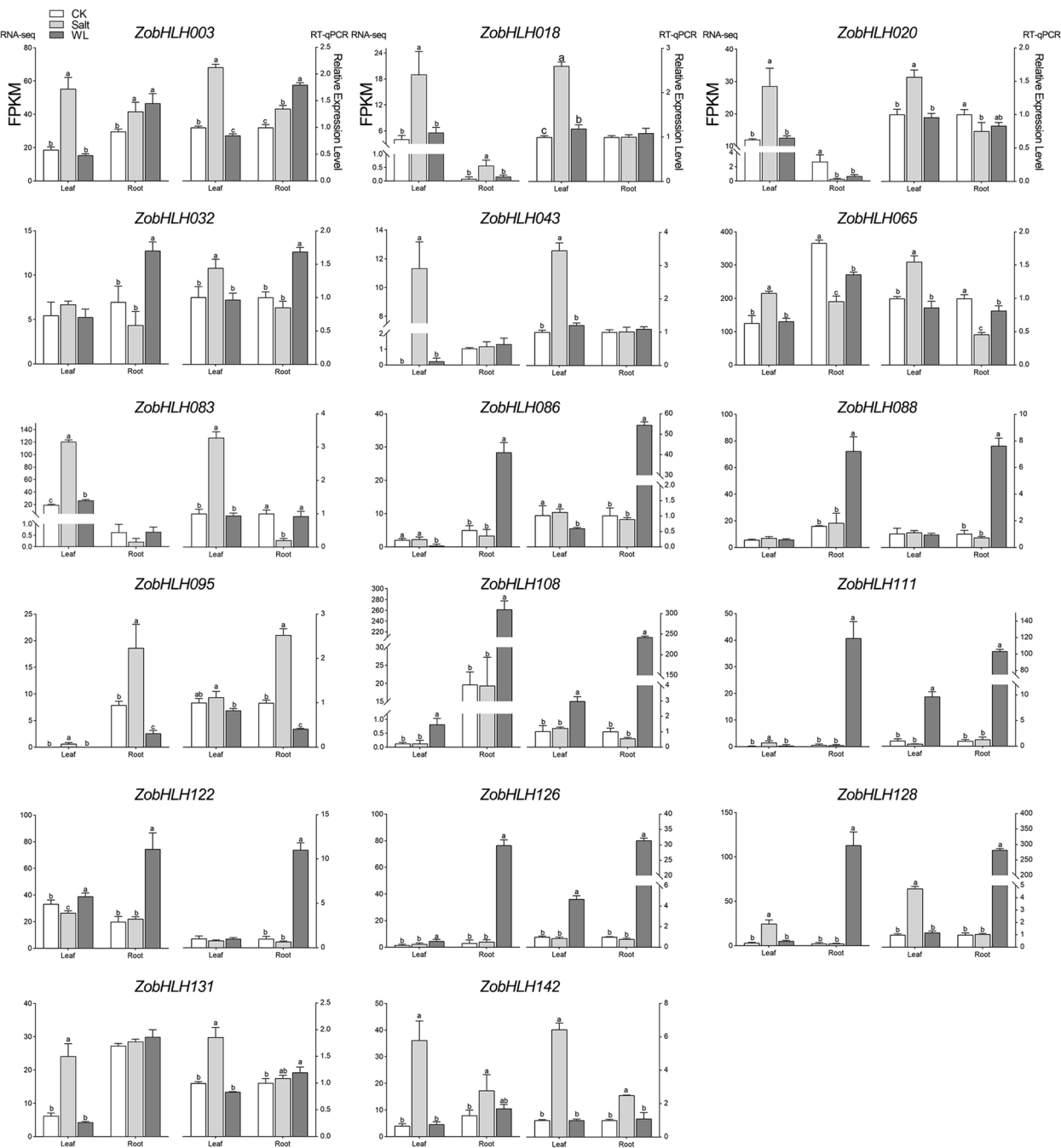


Fig. 9 RT-qPCR analysis of 17 differentially expressed genes. Lowercase letters indicate the difference is significant ($p < 0.05$)

waterlogging stress, the expression levels of *ZobHLH032*, *ZobHLH088*, *ZobHLH122*, *ZobHLH126*, *ZobHLH086*, *ZobHLH111*, *ZobHLH108*, *ZobHLH095*, and *ZobHLH128* were significantly increased. These results were consistent with the RNA-seq data and highlighted the diverse roles of *ZobHLH* genes in response to stresses in ginger.

Subcellular localization of *ZobHLHs* proteins

In this study, we found that *ZobHLH083* and *ZobHLH108* exhibited high expression levels in salt stress and waterlogging, respectively, suggesting that these two *ZobHLH* genes may mediate stress tolerance in ginger. To detect the subcellular localization of *ZobHLH083* and *ZobHLH108* proteins, fusion proteins with GFP tag were constructed and green fluorescence

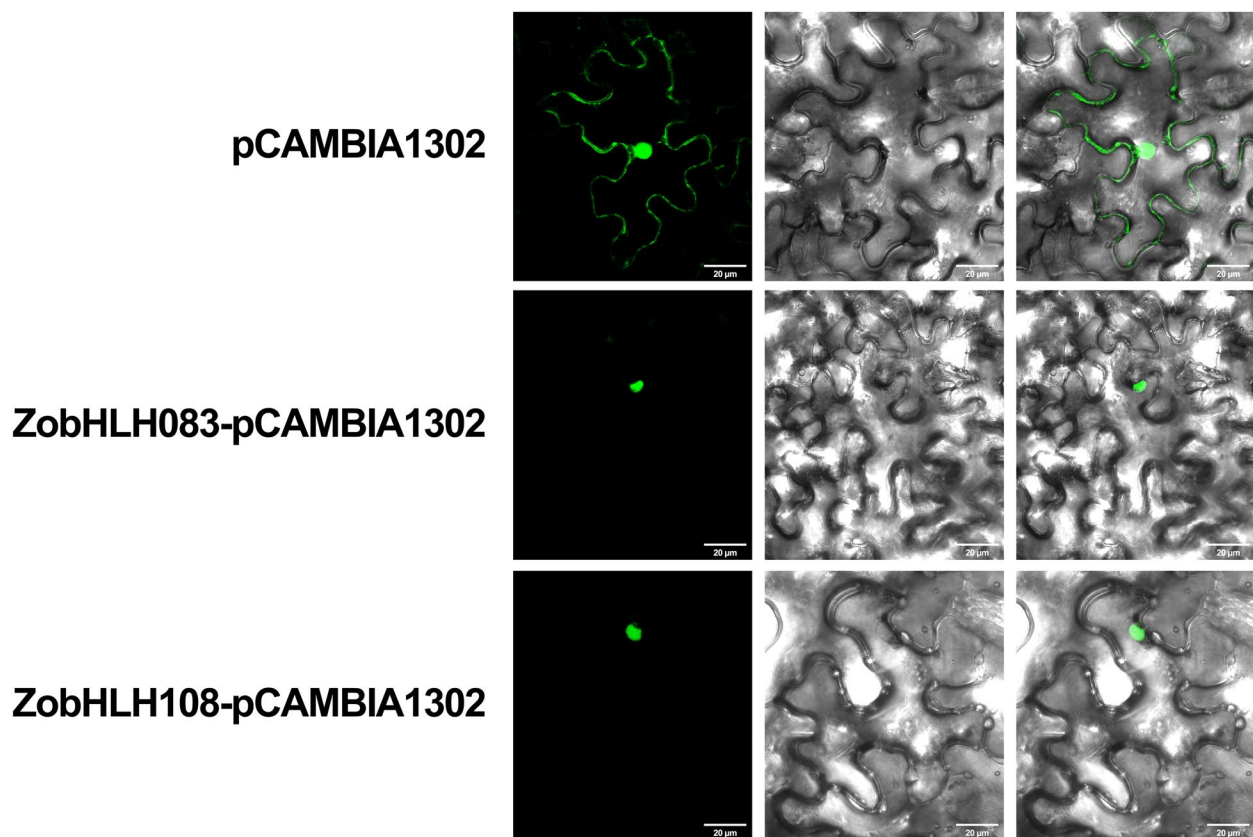


Fig. 10 Subcellular localization of ZobHLH083 and ZobHLH108 proteins. ZobHLHs-GFP were transiently expressed in tobacco cells under CaMV 35 S promoter. From left to right, the pictures show GFP signal in dark, bright, and merged field. Scale bar = 20 µm

was observed. Obviously, the GFP signal of the positive control existed both in the whole cell, whereas ZobHLH083 and ZobHLH108 were mainly localized to the nucleus (Fig. 10). The experimental results were consistent with the predicted results on subcellular localization.

Discussion

The *bHLH* TFs constitute a large family in higher plants, and extensive research has highlighted their crucial roles in plant growth, development, and stress response [42]. Currently, the *bHLH* family has been identified in multiple species including *A. thaliana* [43] and banana [44]. In this study, 142 *bHLH* genes were identified in ginger, which were unevenly distributed across chromosome. The molecular weights of the ZobHLHs proteins identified in ginger ranged from 10615.09 Da to 126549.30 Da, and the isoelectric points ranged from 4.8 to 9.87, which differed from banana, *Loropetalum chinense* and *A. thaliana* [44, 45], reflecting evolutionary diversity and adaptation within the *bHLH* gene family. Protein stability is necessary for life activities and metabolic processes, and unstable proteins may be readily aggregated. While the

unstable proteins in complex tend to be stable means that they should interact with other molecules to be stabilized and then perform functions [46]. In this study, all the ZobHLHs are identified to be unstable proteins. Which suggests that ZobHLH in ginger may function as polymers. Predictions of subcellular localization showed that most ZobHLH proteins are located in the nucleus, which is consistent with previous findings in other species [47, 48].

Based on phylogenetic analyses, ZobHLH proteins can be classified into 15 subfamilies. Conserved motifs and gene exon intron structures of ZobHLH proteins in the same subfamily are similar or identical, suggesting a common origin from ancestral *bHLH* gene. These characteristics in ZobHLH proteins are consistent with previous findings in Arabidopsis, rice and maize [49]. Further analysis showed all the ZobHLHs contain Motif-1 and Motif-2, however, notable differences were observed in conserved motifs between different subfamilies, suggesting evolutionary divergence in both structure and function of ZobHLH proteins. In addition, we identified a diverse range of *cis*-acting elements in the promoter regions of *ZobHLH* genes associated with abiotic

stress responses. The *cis*-acting elements are crucial for transcriptional regulation, influencing diverse biological processes and gene expression [50]. Similarly in rye [51] and maize [52], the bHLH promoter likewise contains a variety of *cis*-regulatory elements related to abiotic stress responses [53].

Gene duplication events are a crucial evolutionary mechanism driving gene expansion in plants [54]. In this study, we identified only one pair (*ZobHLH076*, *ZobHLH077*) of tandemly duplicated genes and 42 pairs of segmentally duplicated genes. The presence of segmental duplications on different chromosomes suggests that segmental duplication significantly contributes to the expansion of the *ZobHLH* gene family. KA/KS analysis of the homologous gene pairs showed that all *ZobHLH* genes underwent purifying selection, indicating a high degree of evolutionary conservation. Duplicated genes may exhibit functionally redundant, highlighting the importance of functional studies to understand their underlying evolutionary mechanisms [55]. Additionally, constructing interspecies of co-variation maps with two dicotyledonous and four monocotyledonous plants revealed that ginger is closely related to monocotyledonous plants and more distant to dicotyledons plants. Since ginger and banana are tropical monocotyledons that evolved from a common ancestor [23], thus they share the highest number of covariates, suggesting that their genes may have similar structural features and functions.

Proteins function through stable or transient complexes, which are crucial for their biological roles in plants [56]. Exploring protein interactions offers valuable insights into the functions of *ZobHLH* genes. In this study, a complex network of *ZobHLH* protein interactions was classified into three tiers based on connectivity. The innermost tier includes the four most interactive proteins (*ZobHLH132*, *ZobHLH103*, *ZobHLH126* and *ZobHLH141*), showing their potential pivotal role in the protein interaction network of ginger. In addition, *ZobHLH132* was identified as the most highly connected protein, which as a homolog of *AtbHLH093* involved in the differentiation of stomatal guard cells [39]. It provides clues to hypothesize that *ZobHLH132* performs a similar function in ginger.

The latest research shows that bHLH TFs can be involved in the regulation of vascular bundle development and rhizome expansion in ginger, and it is well known that ginger rhizome size is a major trait determining the ginger yield [57]. In this study, transcriptomic analysis showed that 95 *ZobHLH* genes were expressed during later stages of rhizome development. Notably, *ZobHLH065* and *ZobHLH064* were highly expressed at all developmental stages. Phylogenetic analysis showed that *ZobHLH065* is closely related to *AT1G32640*

(*MYC2*), *AT5G46760* (*MYC3*), *AT4G17880* (*MYC4*), and *AT5G46830* (*MYC5*), all of which contain the bHLH domain and the bHLH-MYC_N domain [58]. These MYC transcription factors, part of the bHLH MYC subfamily, which interact with Jasmonate Zim-domain proteins and are important regulators of JA signaling branches [59]. While, JA signaling influences various aspects of plant growth and stress responses, including lateral root development and response to adversity stress in plants [60, 61]. That means the rhizome expansion of ginger may be regulated by *ZobHLH065* according to JA signaling pathway. *ZobHLH064* is a homolog of *AT5G54680* (*ILR3*) and *AT1G51070* (*bHLH115*), which are involved in metal homeostasis in *Arabidopsis* growth and development [62]. In addition, eight *ZobHLH* genes containing regulatory elements for flavonoid biosynthesis were identified. Among which six genes (*ZobHLH098*, *ZobHLH078*, *ZobHLH022*, *ZobHLH050*, *ZobHLH117*, *ZobHLH126*, and *ZobHLH069*) were expressed during rhizome expansion. These results suggest that *ZobHLHs* likely play regulatory roles in the ginger development, and especially in the process of rhizome expansion.

In production, ginger belongs to continuous crop, which requires large amounts of chemical fertilizers during cultivation, leading to serious salinization [24], the salt stress can inhibit the growth of ginger seedlings (leaf fresh weight and root fresh weight), which leads to lower yield of ginger [63]. In this study, the expression pattern of *ZobHLHs* under salt stress and waterlogging treatment showed significant fluctuations. For example, under salt stress, *ZobHLH083* was significantly up-regulated in leaves, and phylogenetic analysis showed that this gene was homologous to *ICE*, which was found to enhance cold tolerance, osmotic and salt tolerance of plants, et al., and play a vitally important transcriptional regulatory role in abiotic stresses [64]. In addition, *ZobHLH142* and *ZobHLH003* were up-regulated in response to salt stress in both leaves and roots. *ZobHLH142* is a homolog of *AT4G36930* reported to respond to abiotic stresses in cucumber by regulating cell signaling [65]. *ZobHLH003* is a homolog of *AT4G16430* involved in balancing plant defense mechanisms against external stresses with [66]. These results suggest that *ZobHLHs* significantly function in ginger resistance to stresses.

Ginger root system is undeveloped, shallow growth, moisture-loving but intolerance of waterlogging. It is very susceptible to waterlogging, which could seriously affect ginger yield [29]. Waterlogging stress inhibits aerobic respiration and impedes plant growth and development [67]. In this study, a large number of DEGs were identified in the transcriptome of waterlogging stress. For example, *ZobHLH108* is homologous to *AT2G28160* (*FIT1*) and significantly induced under waterlogging stress. Studies

have shown that *FIT1* which is expressed mainly in the root system [68] can integrate iron deficiency and oxidative stress responses, thereby promoting normal plant growth and development [69]. Therefore, we hypothesized that *ZobHLH108* perform a similar function under waterlogging. Notably, *ZobHLH128* was significantly up-regulated in both salt-stressed leaves and roots under waterlogging treatment compared with the control, highlighting its crucial role in abiotic stress response in ginger. Studies showed that *NtbHLH* genes are rapidly induced when tobacco subjected to waterlogging stress [70], *MfbHLH38* acts positively on plant defense through the ABA pathway and enhances the ROS (reactive oxygen species) scavenging system to reduce oxidative damage [71]. *AtbHLH112* responds to abiotic stresses by regulating the expression level of *proline dehydrogenase 1* to increase the proline content, and then decrease ROS accumulation and water loss [72]. In conclusion, *ZobHLH* transcription factors have critical functions in the response to abiotic stresses including salt and waterlogging [73], and *ZobHLH083* and *ZobHLH108* are potential genes during abiotic stress resistance in ginger. However, research is still needed for further validation.

Conclusion

In this study, 142 ginger *ZobHLH* genes were comprehensively identified. Bioinformatics analysis revealed their physicochemical properties, gene structures, and expression patterns. The findings indicated that *ZobHLH* genes play crucial roles in ginger rhizome expansion and in responding to salt and waterlogging stresses, the expression level of *ZobHLH065* was highest in the middle and late stages of rhizome expansion, and *ZobHLH083* and *ZobHLH108* were significantly up-regulated under salt and waterlogging stresses, respectively. These results provided valuable genetic resources and theoretical foundations for molecular breeding and genetic improvement of ginger. The biological functions of these genes and their synergistic effects on enhancing the resistance to environmental stresses in ginger need to be further studied.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12864-025-11284-8>.

Supplementary Material 1.

Authors' contributions

Y.L. and X.Z. conceived and designed the experiment. D.L. conducted experiments and wrote the manuscript. P.Z. and M.Y. participated in samples collection and experiment conduction. T.Z. and Y.W. performed data analysis. D.L. conducted molecular experiments and X.Z. directed the experiment. X.Z. and Y.L. finalized the manuscript for submission. All authors read and approved the manuscript.

Funding

This study was financially supported by Open Fund Project of Engineering Research Center of Ecology and Agricultural Use of Wetland, Ministry of Education, Yangtze University (KF202312), Hubei international science and technology cooperation project (2024EHA011), the Hubei Provincial Key R&D Initiative Projects (2022BBA0061, 2021BBA096) and Chongqing Yingcai Excellent Science Project (2022CQYC027).

Data availability

Ginger rhizome expansion transcriptome data are deposited in the NCBI database (Accession Number: PRJNA788194), and additional data is provided within the manuscript and supplementary information files.

Declarations

Ethics approval and consent to participate

We confirm all the experimental research on ginger, including the collection of plant material, were performed in accordance with relevant institutional, national, and international guidelines and legislation. The ginger (*Zingiber officinale* Roscoe) plants used in this experiment are authorized by the Spice Crops Research Institute, College of Horticulture and Gardening, Yangtze University.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 2 October 2024 Accepted: 23 January 2025

Published online: 13 February 2025

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